

Mass and Weight

History

The concepts of mass and weight were revolutionised by **Sir Isaac Newton** in the 17th century.

Before Newton, people used the terms "mass" and "weight" interchangeably. In everyday life, this still happens — we say "this weighs 5 kg" when we really mean "its mass is 5 kg".

Newton's Second Law () provided the first clear distinction:

- **Mass** is a measure of the amount of matter in an object — it resists changes in motion (inertia)
- **Weight** is the gravitational force acting on that mass

The kilogram was originally defined (1795) as the mass of one litre of water. In 1889, it was replaced by a platinum-iridium cylinder — "Le Grand K". In **2019**, the kilogram was redefined in terms of Planck's constant (), linking it to fundamental physics instead of a physical object.

Myth vs Fact: An astronaut in space is NOT "weightless" because they have no mass. They still have mass — they are weightless because they are in **free fall** around Earth. The ISS and everything inside it is falling towards Earth constantly, but moving sideways so fast they keep missing it!

$$F = ma$$

Nature & Applications

- 🌕 **Why does the Moon orbit Earth?** Gravity provides the centripetal force
- ⚖️ **Spring balances** measure weight (force), not mass
- 📱 **Digital scales** typically measure the force and divide by g to display mass
- 🚀 **Rocket launches** must account for the changing weight as fuel burns and gravity decreases with height
- 🌌 **On different planets**, your mass stays the same but your weight changes dramatically

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Theory

Mass

- **Definition:** The amount of matter in an object
- **SI unit:** kilogram (kg)
- **Scalar** quantity (has magnitude only, no direction)
- **Does NOT change** with location — your mass is the same on Earth, the Moon, or in deep space
- **Measured with:** a top-pan balance or a beam balance (comparison against known masses)

Weight

- **Definition:** The gravitational force exerted on an object
- **SI unit:** newton (N)
- **Vector** quantity (direction: towards the centre of the gravitating body)
- **DOES change** with location — your weight on the Moon is about 1/6 of your weight on Earth
- **Measured with:** a spring balance (or newton-meter / force meter)

The Equation

Where:

- = weight (N)
- = mass (kg)
- = gravitational field strength (N/kg)

On Earth, N/kg. **This is the same number** as the acceleration due to gravity (m/s^2) — because:

Gravitational Field Strength

Celestial Body	(N/kg)
Earth	9.8
Moon	1.6
Mars	3.7
Jupiter	24.8
Sun	274
Deep space	0 (weightless)

Difference Between Mass and Weight

Property	Mass	Weight
Definition	Amount of matter	Gravitational force
SI unit	kilogram (kg)	newton (N)
Quantity type	Scalar	Vector
Changes with location?	No	Yes
Measured by	Balance	Spring balance
Symbol		

Inertia

Inertia is the tendency of an object to resist changes in its motion. It is directly proportional to mass:

- A heavy truck has **more inertia** — harder to start moving, harder to stop
- A tennis ball has **less inertia** — easy to throw, easy to catch

$$W = mg$$

$$1 \text{ N/kg} = 1 \text{ m/s}^2$$

$$F = ma \quad (\text{more mass} \rightarrow \text{more force needed to accelerate})$$

$$\frac{1}{6} W \quad m \quad g \quad g \approx 9.8 \quad 9.8 \quad g \quad m \quad W$$

Worked Examples

Example 1: Weight on Different Planets

An astronaut has a mass of 75 kg.

- (a) What is their weight on Earth (N/kg)?
- (b) What is their weight on the Moon (N/kg)?
- (c) What is their mass on the Moon?

Solution (a):

Solution (b):

Solution (c):

Mass does not change — still **75 kg** on the Moon.

Example 2: Finding Mass from Weight

A bag of rice has a weight of 49 N on Earth. What is its mass?

Solution:

Example 3: Comparing Inertia

Two identical trolleys are pushed with the same force. Trolley A has a 2 kg mass on top, Trolley B has a 5 kg mass. Which accelerates more slowly? Explain.

Solution:

Trolley B accelerates more slowly because it has greater mass, therefore greater inertia. From , for the same force, a larger mass gives a smaller acceleration.

$$W = mg = 75 \times 9.8 = 735 \text{ N}$$

$$W = mg = 75 \times 1.6 = 120 \text{ N}$$

$$m = \frac{W}{g} = \frac{49}{9.8} = 5.0 \text{ kg}$$

$$g = 9.8 \quad g = 1.6 \quad a = \frac{F}{m}$$